2024A7PS0002G
AI Lab1: Neural Models

1. The input space consists of 2 sensors, x1 and x2. These can take values 0 and 1. The output (y) should be set to 1 when the sensors show different values and 0 when they “agree” ie. show the same value.
Since the “positive” and “negative” examples lie on diagonals of the square representation of the input space, a linear model cannot give a correct output. (since no line can be drawn on this space to divide it into distinct true/false regions)

2. Model design:
- 2,2,1 neural network with 2 input neurons, 2 hidden neurons, 1 output neuron
- Nonlinear activation function: sigmoid
- Binary cross entropy loss + gradient-based optimiser

Validation Criteria:
- Final loss is lower than the initial loss
- All inputs are classified correctly
- The gradients of the model parameters are nonzero during training

3. return a python file for the (problem statement) that builds a model with the following specifications (image of above) After training, report the final loss, all four probabilities, thresholded labels, and one parameter-gradient tensor. Set a random seed for reproducibility and explain each test in one sentence

4. Code included in python file

5. From the code output:

```
Initial loss: 0.70276976
Final loss: 0.00002628
Probabilities: [3.949524398194626e-05, 0.9999759197235107, 0.9999761581420898, 1.7779178961063735e-05]
Thresholded labels: [0, 1, 1, 0]
First-layer weight gradient:
 tensor([[-3.6753e-08, -2.8846e-08],
        [-2.8643e-07, -3.2674e-07]])
Loss test: PASS - the final loss should be lower than the initial loss.
Classification test: PASS - all four thresholded predictions should match the XOR targets.
Gradient test: PASS - a nonzero parameter gradient shows that backpropagation produced an update signal.
```

6.1) We need nonlinear activation functions between layers to achieve a nonlinear output, as adding only linear operations to the input can only result in a linear output.

6.2) The useful learning signal was shown by the fact that the loss decreased from about 0.70277 to 0.000026, and the final predictions were [0, 1, 1, 0], correctly classifying all four XOR examples. Hence, the gradients were not just nonzero as they caused parameter updates that improved the model.

6.3) If the two hidden units start with identical weights, they will produce the same outputs and receive the same gradients during backpropagation. Their weights can only remain identical, so the two neurons will not learn different features.

6.4) Scientific: depends on derivatives of each function.
Engineering: could not understand how the activation function changed the output.

6.5) The output layer determines how the model represents its prediction, and the loss defines how that prediction is compared with the target.

6.6) The LLM helped me understand concepts and generate code with my specifications. I had to make sure that the code actually aligns with the problem statement and verify the outputs.

6.7) We can keep tests such as monitoring training and validation loss, checking predictions on a validation set, inspecting gradient norms, and checking for vanishing or exploding gradients. Tests that require performing exhaustive gradient checks would become too expensive for a large model because of the larger number of parameters and training examples.